Phantasm player's manual

Phantasm turns the Ensoniq Mirage into a wavetable synthesizer. This manual is for the person playing it: it says what each control does, in the order you will need it, and what to do when something goes wrong. Every number in it is a number the front panel shows.

The printed parameter card is the one-page version of section 11. The sections run from what the instrument is (1), through switching it on (2) and the panel (3), to the sound (5 to 9), the level of the sound (10), the reference (11), MIDI (12) and what the display is telling you (13). If you are new to wavetable synthesis, read the quick-start guide, docs/QUICKSTART.md, first: it explains tables, scanning and the voice in pictures.

1. What Phantasm is

Phantasm is an operating system for the Mirage. It replaces the sampler with a wavetable synthesizer and leaves the instrument's own sound hardware as it was: the Ensoniq sound chip plays the waves, and the analog filters shape them.

- **Waves.** A wave is one cycle of a sound, 256 points long. The disk holds 512 of them. The first 256 are the factory's: analog shapes, harmonics, voices, bells and so on. The next 128 are noise. The last 128 are the user area, for waves of your own (section 5).
- **Tables.** A wavetable is an ordered list of waves. A patch plays one table, and moves along it as you play: with the envelope, the LFO, the keys and the wheels (section 5).
- **Voices.** Eight voices, each with two oscillators, A and B, one filter and its own envelopes. Poly, mono and unison modes use them differently (section 9).
- **Envelopes.** Three: ENV1 shapes the loudness, ENV2 the filter, and ENV3 is free to move the pitch, the resonance, the LFO or the wave position (section 8).
- **Patches.** A hundred, numbered 00 to 99, kept on the disk, one to a sector. Everything about a sound is in its patch (section 4).

What it does not have: there is no split or layer (a patch is one sound) and no LFO sync to MIDI clock. A parameter changed on the panel is heard at once and lost when another patch is loaded, unless you save it first.

2. Getting started

What you need

- The Mirage, a Gotek floppy emulator in place of its drive, and a USB stick holding the Phantasm disk image, which is a file ending in .hfe.
- Give the file a short name. The Gotek's small display shortens long ones.
- For notes, a MIDI keyboard on the Mirage's MIDI input, or the Mirage's own keyboard. The Mirage's own wheels may be noisy on an old unit; a MIDI controller's bend and mod wheels are the better test.

Switching on

1. Put the stick in the Gotek and select the Phantasm image on its display.
2. Switch the Mirage on.

3. Wait. The disk's waves are read into the instrument before it will play, so boot takes about 21 seconds and there is little to see on the display until it ends. This is not a fault.
4. When boot ends the display shows the version, two digits: 12 for version 1.2 (a release build shows 1.2 with a dot between them). It stays until you touch the panel.

Patch 00 is loaded at boot, and the MIDI channel, omni, master tune and transpose are restored from the disk if you saved them (section 4).

Is it working?

Play a note. If you hear it, the disk, the waves and the voices are working. If the keys are silent, send a note from a MIDI keyboard on the channel parameter 95 shows, or check that omni (96) is 1.

If the display shows a letter E and a character at boot or after a disk operation, the disk did not read: see section 13. The instrument cannot be damaged by a bad disk. It never writes to the disk except when you save.

3. The front panel

Phantasm keeps the Mirage's own way of editing: every sound control is a numbered parameter, a two-digit number from 10 to 98.

Reading and changing a parameter

1. Press **PARAM** and type the two digits of the parameter number, for example 7 0 for the filter cutoff. The display shows the parameter's current value.
2. To set a value, press **VALUE** and type two digits: 0 4 sets four. A value is always typed as two digits. It takes effect when the second digit goes in: there is no Enter to press.
3. Or nudge the value with **On/Up** and **Off/Down**, one step at a time. They stop at the top and bottom of the parameter's range.
4. **Cancel** abandons a number or a value you have started and has not yet taken effect.

A typed value larger than the parameter's range is clamped to the top of it. A parameter number that does not exist shows EE.

An edit goes straight into the sound, so you hear it at the next control step, or at the next note for some parameters (the manual says which). There is no undo: to go back to the saved sound, load the patch again (section 4).

Actions

Two numbers are not parameters but one-shot actions. Press **PARAM**, type the two digits, and the action runs when the second digit goes in.

Action	What it does
94	Restarts the wave sequence at its first step (section 5)
93	Saves the user waves that have changed (section 5)

The patch buttons

Four of the Mirage's own buttons choose and save patches: LOAD Upper, LOAD Lower, SEQ Load and SEQ Save. If your unit's panel does not have them, use program change and the editor (section 12).

- **LOAD Upper** loads the next patch and **LOAD Lower** the previous one. Both wrap round: Upper from 99 goes to 00.
- **SEQ Load**, then two digits, loads that patch. It loads on the second digit.
- **SEQ Save** then **Enter** saves over the patch you are on. **SEQ Save** then two digits saves a copy under that number (section 4).

While the disk is working the display shows Ld (load) or Sd (save), and then the patch number, or an error (section 13). Keys and MIDI notes sent while the disk is working are lost, and every voice is released when it ends. Wait for the number before you play.

4. Patches

A patch is one sound: its wavetable, sequence, envelopes, LFO, filter, oscillators and modes, kept as one sector on the disk. There are 100, numbered 00 to 99. The factory disk has eight, 00 to 07; 08 to 99 are empty starting points.

Loading

Load with LOAD Upper, LOAD Lower or SEQ Load and two digits (section 3), or by MIDI program change: program p loads patch p, for p up to 99. Programs 100 to 127 are ignored.

A load takes about two seconds. Loading the patch you are already on loads it again, which is how you throw away edits.

Unsaved edits are discarded without asking when another patch loads, from the panel, a program change or a MIDI editor.

Saving

1. Make the sound you want with PARAM and VALUE.
2. Press **SEQ Save**. The display shows the number of the patch you are on.
3. Press **Enter** to overwrite that patch. Or type two digits to save a copy as that number. There is no prompt, so a number that already holds a patch is overwritten.
4. The display shows Sd while the disk works, then the number. After a save-as, the number is the copy: it is now the patch you are editing.

Cancel before the second digit abandons the save. The save is written and then read back from the disk, and an error shows if it did not match.

Saving the instrument settings

The MIDI channel (95), omni (96), master tune (97) and transpose (98) belong to the instrument, not to a patch. They are written to the disk together with a patch save, but only when one of them changed since boot. Changing patches never changes them. A disk saved before master tune and transpose existed loads with both at 50 (none).

Notes

- A save is refused (EP) when the disk is write protected (section 13).
- After a failed load the instrument plays a plain default sound and refuses to save until a patch loads well, so a read error cannot overwrite a good patch with the default. Load a patch that works, then save.

- The Gotek has one image at a time. To work with more than 100 patches, swap images.

5. Waves, tables and sequences

The wave bank and the tables

The factory disk's waves 0 to 255 are made from recipes: analog sweeps, harmonic series, vocal shapes, bells, FM, and so on. A patch does not choose waves one by one. It chooses a wavetable with parameter **10**, which is the table's number in the patch's own list, starting at 0, and the table is an ordered list of waves.

Parameter **12**, wave position, is where in the table a note starts, as a share of the table: 0 is the first entry and 99 the last, whatever the table's length, and 50 is the middle. On a table of up to 87 entries every entry has a value of its own. On a longer one a value now and then steps over an entry (28 of the 128 on the longest), and only a patch from the editor or a bank file can start on those. Choosing another table with 10 keeps the start at the same share of the new one.

Changing 12 moves the notes that are sounding, whether it comes from the panel, a controller (CC 12) or the editor: each moves to the new start at once, and keeps its own sequence step and its own velocity and key offsets. Loading a patch does not move them. A patch saved by an earlier release stored an entry number here, so it now starts nearer the beginning of its table: set 12 again.

Modulation then moves the position away from where the note started:

Source	Parameter
Velocity	14
Key, higher notes further	15
ENV3	85 set to 5, and its depth 86
LFO	17
Mod wheel	38 set to 1, and its depth 36
Channel pressure	39 set to 1, and its depth 37

Parameter **13**, wave wrap: 0 holds the position at the end of the table, 1 wraps it round to the other end.

A scan switches waves directly, with no crossfade, so you hear the steps between waves. That is the instrument's sound, not a fault.

Wave sequences

A **sequence** is a list of positions in the table. Parameter **11** chooses one, and parameter **18** says how it is played, on each note:

18	Mode	What happens
0	Off	Each note starts at the position in 12
1	Step	Each note takes the next step in the sequence
2	Per voice	Each voice has its own place, going round the eight
3	Random	Each note picks a step at random
4	LFO	Each voice takes the next step every LFO cycle while its key is held

The factory palette's sequences, which parameter 11 picks between (turn 11 with a mode of 1 or more and the next note starts elsewhere):

11	Name	Steps
0	eight	positions 0 to 7
1	four	0, 8, 16, 24
2	spread	eight positions chosen to sound as unlike as possible on the vocal table
3	hold	position 0 only: plays as if the mode were 0
4	flip	the two ends of the table, 0 and 31
5	leap	eight positions jumping between the start and the end
6	scatter	eight positions across the table

The last four are the *glitch* sequences: every jump lands on a wave that sounds clearly unlike the last one, on every factory table. Patches 00 to 07 are saved with a mode of 1 or more, on `hold` except patches 02, 03 and 06, so 11 does nothing until you turn it. Patch 06 plays `scatter` in mode 4, the wave jumping about nine times a second while a key is held. The patch on the user palette (07) has `eight`, `four` and `hold`.

The sequence chooses where a note starts, and the modulation above moves it from there. A strong scan (ENV3 or the LFO to the wave pointer) moves the wave further than the sequence's offsets, so turn that down to hear them. Parameter **19** restarts the sequence after that much silence between notes (0 is never). Action **94** restarts it by hand.

In mode 4 a note starts where mode 2 would put it, on its voice's own step, and from then on the step moves on by one each time the voice's LFO completes a cycle, wrapping from the last entry to the first. Each voice steps on its own LFO: with LFO sync (33) at 1 the voices of a chord step together, and at 0 they drift apart. The LFO rate (parameter 30, 0 holds the step) sets how fast; the LFO delay (parameter 35) holds the first step until it has run out; and the step stops when the key is let go. The LFO's depths (17, 31, 34) are separate: at 17 = 0 the LFO only steps the sequence, and above 0 it also sweeps within each step. The random LFO shape steps at the LFO rate too.

The user area

Waves 384 to 511 are 128 places for waves of your own. The factory disk puts 32 waves there: Evolver single-cycle waves in places 384 to 415, played by patch 07's table, named evolver. The other 96 are empty (a flat line).

To put a wave of your own there you send it from a computer as a MIDI sample dump (section 12). Then you must save it, or the next boot loads the disk's waves over it.

1. Send the dump (section 12). Notes are ignored while it arrives, and the keys work again when it ends.
2. Press **PARAM 9 3**. The display shows Sd while the changed waves are written to the disk, then the patch number.
3. If nothing has changed, 93 writes nothing and the display just shows the patch number.

Only the changed waves are written. Each is read back and compared, and if one does not match the display shows Er, and that one and the ones after it are tried again at the next save (section 13).

A received wave is played only if a table names it. Patch 07's table, evolver, names 384 to 415, so a dump to sample numbers 0 to 31 replaces waves that table plays, and a dump to 32 and up is stored but no factory table plays it. A MIDI editor can write a patch with a table that names them.

The noise region

Waves 256 to 383 are noise, not waves to scan. They are played by oscillator B only, as its source (section 6).

6. Oscillators A and B, and noise

Each voice has two oscillators. A plays the wave the table chooses. B plays the same wave, a little different, or it plays noise.

Tuning

Parameter	Controls	Shows	Step
20	Coarse A	38 to 62	one semitone
21	Fine A	29 to 71	4.7 cents, about a semitone each way in all
22	Coarse B	38 to 62	one semitone
23	Fine B	29 to 71	4.7 cents

For all four, **50 is no shift**. 38 is a whole octave down and 62 a whole octave up. A fine step is a 256th of an octave, not a cent, so neighbouring numbers are always different pitches. Tuning is applied at the next note.

25 sets the bend range, in whole semitones from 0 to 12, for the full travel of the wheel. 2 is a whole tone and 12 an octave.

Levels

27 is A's level (99 is full, 0 is off) and **24** is B's (0 is off). B is taken from the same level as A, not from what is left, so turning B up makes the voice louder (section 10).

Noise and sync on B

28 chooses what B plays:

28	B plays
0	The same wave as A
1	White noise, following the key
2	Pink noise, following the key
3	Scrape noise, following the key
4	White noise at a fixed rate
5	Pink noise at a fixed rate
6	Scrape noise at a fixed rate
7	The same wave as A, hard-synced to A

White is plain hiss and pink a darker one; scrape is metallic, noise ringing through the modes of a struck bar, like metal being scraped. In 1 to 3 the noise's pitch follows the key, so it gets higher up the keyboard; in 4 to 6 the rate is the same for every key, that of C4, and 22 and 23 still shift it. Pink and scrape take on a pitch towards the top of the keyboard: that is the instrument's lo-fi character.

For noise on its own, set **27** to 0, so A is silent, and **24** to the level you want. The filter and envelopes act on noise as on a wave.

Sync (7). B plays A's wave and is reset every time A's wave starts over, so what you hear sits at A's pitch and **22** and **23** change the timbre, not the pitch: sweeping 22 gives the classic sync sound. The pitch you hear is A's, so a key played on A and B in unison is one note. B's level (24) and A's level (27) still act, and the envelopes are as for any wave; set 27 to 0 to hear the synced B on its own. The sync is set when a note starts, so a change to 28 is heard at the next note, and a patch with 28 at 0 to 6 sounds as it did.

Vintage

26 makes each note drift: at every note, each voice gets a small pitch and filter offset of its own, held for the note. 0 is off, and at 99 the pitch spread is about plus or minus 38 cents.

7. Filter and amp

The filter

Every voice has its own low-pass filter, and the voices are added after their filters. These parameters set it:

Parameter	Controls
70	Cutoff
71	Resonance
72	How far ENV2 moves the cutoff
45	How far a soft key takes that away
73	How far the key moves the cutoff
74	How far velocity moves the cutoff
34	How far the LFO moves the cutoff
77	How far the wave pointer's place in the table moves the cutoff
37	How far channel pressure moves the cutoff, when 39 is 2

Parameter 77 is the one to reach for when a sound scans a wavetable. A table that runs dull to bright puts its energy higher the further up it you go, and a cutoff that stays where it was then takes away what the scan added: the bright end comes out quiet. At 77 = 50 the pointer does nothing to the filter. Above 50 the filter opens as the pointer climbs the table, 0 at the first entry and the full amount at the last, on a table of 32 entries or of 128 alike, and below 50 it closes. At 99 the last entry is 98 steps of cutoff above where it would be, nearly three and a half octaves, and 82 is about 64 steps. The amount follows where the pointer is when it sounds, after the sequence, the LFO, ENV3 and the wheel have all moved it, and oscillator B follows the pointer, so it has no setting of its own. A change to 77 takes effect on the notes already sounding.

Parameter 45 makes the filter envelope follow how hard the key is struck. At 0, 72 is the depth whatever the velocity. At 99 a key struck at full velocity gets all of 72, and a softer one the same fraction of it as its velocity is of the top, so a note at velocity 20 gets about a sixth: a hard key opens the filter and a soft one barely moves it. In between, 45 at 50 takes off half as much. Only the envelope's depth is scaled; the cutoff itself (70) and the velocity term (74) are as they were. Velocity here is the key's own, so the notes of a mono legato phrase keep the depth of the note they were slurred onto, and all the voices of a unison note share one. A change to 45 or to 72 takes effect on the notes already sounding.

Parameter 37 with 39 at 2 opens the filter from the pressure of a MIDI keyboard's aftertouch. At 37 = 99 and full pressure it adds 253 steps of cutoff, more than the filter's whole range, so a low cutoff opens

fully, and 37 at 50 gets about half as far; with no pressure it adds nothing, whatever 37 is. A note already sounding follows the pressure at once, and a change to 37 or 39 takes effect within a moment.

Resonance can be raised until the filter rings on its own, and it also raises the level of the sound around the cutoff (section 10).

The amp

There is no separate amplifier circuit. The loudness is set by the sound chip's volume, shaped by ENV1.

Parameter	Controls
75	Amp level: the loudest a voice gets
76	Velocity to amp: how much harder keys are louder
50 to 54	ENV1, the amp envelope (section 8)

At 76 = 0 every key sounds at the level 75 sets. A change to 75 or 76 takes effect at the next note you play. Section 10 is the guide to setting them.

8. Modulation

Envelopes

There are three envelopes, and they have the same five parameters in a row: the first digit is the envelope, the second the stage.

Envelope	Used for	Parameters
ENV1	Loudness	50 attack, 51 peak, 52 decay, 53 sustain, 54 release
ENV2	Filter cutoff, by 72	40 attack, 41 peak, 42 decay, 43 sustain, 44 release
ENV3	Free: the destination in 85, depth in 86	80 attack, 81 peak, 82 decay, 83 sustain, 84 release

Attack, decay and release are times: higher numbers are slower. The peak is the level the attack reaches, and the sustain is the level held while the key is down. ENV1's decay and release are exponential and shown 0 to 85.

Key speed (47, 57 and 87) makes an envelope faster on higher notes. 50 is none, and 99 halves the decay and release for every octave above C4 (key 60).

ENV3's destination (85):

85	ENV3 moves
0	The pitch of both oscillators, A and B
1	The pitch of oscillator A only
2	The pitch of oscillator B only
3	The resonance
4	The LFO's depth
5	The wave position

With B source 7 (hard sync, 28), 2 sweeps the synced oscillator against a fixed master, the classic sync sound, and 1 sweeps the master against a fixed slave. At 0 the two move together and the ratio between them stays fixed, so the sweep is a plain transposition. With B off (the B level at 0), 1 sounds as 0 and 2 does nothing.

The depth, **86**, runs both ways: 50 is none, and below 50 reverses the movement.

ENV3 loop (88) makes ENV3 a second, drawn LFO. At 0 it runs attack, peak, decay and rests at its sustain (83), as it always has. At 1, when the decay has fallen to the sustain level and the key is still down, it climbs again from there, so the attack, peak, decay and sustain level (80-83) are the four corners of a cycle: sustain is the cycle's floor, and with the sustain at 0 the loop sweeps the whole range. Key speed (87) shortens the looped decay up the keyboard like any decay. Lifting the key releases from wherever the loop is. A mono legato phrase keeps the loop going as the notes are slurred; a new, detached note restarts it from nothing. The voices of a unison note loop together. The setting is read at the end of each decay: changing 88 to 1 under a note whose ENV3 is already resting takes effect from the next note. 88 has no MIDI controller number; set it from the panel or by SysEx.

The LFO

Parameter	Controls
30	Rate
31	How far it moves the pitch
34	How far it moves the cutoff
17	How far it moves the wave position
32	Shape: 0 triangle, 1 ramp up, 2 ramp down, 3 square, 4 random
33	Sync: 0 free-running (each voice's LFO keeps going between notes), 1 restart at each note, so the same sweep every time the key is pressed; 2 is reserved for MIDI clock and runs as 0
35	Delay: how long before it comes in after a note starts. At 0 the LFO starts at once, at full depth; above 0 it is silent for the delay and then fades in

Velocity, key and wheels

- **Velocity** moves the wave position (14), the cutoff (74) and the loudness (76), and can scale how far ENV2 moves the cutoff (45, section 7).
- **Key** moves the wave position (15) and the cutoff (73).
- **Pitch bend** moves every voice, by the range set in 25.
- **The mod wheel** (CC 1) has its own depth, **36**, and a destination, **38**: 0 raises the LFO's depth, and 1 moves the wave position.
- **Channel pressure** (aftertouch, from a MIDI keyboard that sends it) has its own depth, **37**, and a destination, **39**: 0 raises the LFO's depth as the wheel's 0 does, 1 moves the wave position, 2 opens the filter. It adds to the mod wheel's, so the wheel can be on the LFO and pressure on the wave. 37 at 0 is off. The Mirage's own keyboard does not send pressure. A patch starts with none held.
- **The sustain pedal** holds notes, from the Mirage's pedal or CC 64.

9. Voice modes

60 sets how the eight voices are used.

60	Mode	What happens
0	Poly	Up to eight notes at once, oldest note taken first when they run out
1	Mono, retrigger	One note at a time, the last key played; every new key restarts the envelopes
2	Mono, legato	As 1, but a key played while another is held changes the pitch only and does not restart the envelopes

3	Chord, retrigger	As 1, but every key plays the patch's stored chord built on it
4	Chord, legato	As 2, but every key plays the patch's stored chord built on it

In mono and chord, letting go of the sounding key goes back to the newest key still held.

Chords

In chord mode (3 or 4) each key you play sounds a whole chord on that key, one chord at a time. The patch stores the chord as up to seven intervals above the key you play, so the chord moves with your hand: store C, D#, G and F plays F, G#, C. Every note of the chord is one voice at the key's velocity, so a chord of up to eight notes uses up to all eight voices. Unison and spread are not used in chord mode; they come back when you return to 0-2. A chord note higher than the top C (key 96) is left out. In legato (4) a key played while another is held moves the whole chord to the new key without restarting it, and with glide on the chord slides there in one piece.

To capture a chord:

1. Set **60** to 3 or 4, from the panel, with CC 60 or over SysEx. Both digits start blinking: the Mirage is waiting for a chord. Entering the value 60 already has works too, which is how you capture a new chord.
2. Hold the chord. While the digits blink, each key plays one note as in poly, so you hear what you are building. The lowest key you hold is the root.
3. Let go of any key. The chord is stored, the blinking stops, and from the next key you play, every key sounds that chord.

Keys held with the sustain pedal count as held, so with the pedal down the chord is stored when you lift the pedal. With more than eight keys held, the lowest eight are kept.

To clear the chord, capture a single key: arm, play one key, let go. Chord mode then plays one note at a time, as mono does.

To cancel while the digits blink, set 60 to 0, 1 or 2: the stored chord is kept. Loading a patch also cancels. The chord is part of the patch: save the patch to keep it, and leaving chord mode does not clear it.

Unison, 61, makes one key sound several voices, 1 to 8: 1 is off. The voices are spread in pitch by **62**, in steps of 4.7 cents, from 0 to 21. Eight voices on one key leaves none for chords, because every key uses its own voices. Unison is louder than a single voice, so see section 10.

Glide, 63, slides the pitch to a new note. 0 is off, and 1 to 99 are slower, from an octave in 22 milliseconds to an octave in 33 seconds. In mono, glide happens on a legato slur. In poly, every note glides from the pitch its voice last played.

Vintage, 26, is described in section 6.

Mono and unison combine: the mono note is then the group of unison voices. Voice mode, unison, spread, glide, vintage and the stored chord are part of the patch, so a mono bass, a unison lead or a chord pad is back as you saved it when you load it.

10. Gain staging

Gain staging is setting how loud each part of the sound is, so that the loudest thing you will play comes out clean. If a patch sounds fine on single notes and breaks up when you play chords, turn on unison, or add the second oscillator, the patch is too loud for that material. This section shows where the level

comes from and which control to turn.

The level chain, in order

1. **Amp level (75)** sets a voice's loudest level. Every factory patch uses the same value, which the display shows as 75.
2. **Velocity > amp (76)** lets how hard you play change that level. At 0 every key sounds at the level 75 sets. Raised, a softer key sounds quieter, and a full-force key stays at the level 75 sets. Velocity never makes a note louder than 75 does.
3. **A level (27)** and **B level (24)** then set how much of that level goes to each oscillator. A level at 99 passes all of it. B level at 0 turns B off, and at 99 gives B about as much as A. B is taken from the same level as A, not from what is left, so turning B on makes the voice louder.
4. **ENV1 (50 to 54)** shapes the level over the life of the note.
5. **The filter** (cutoff 70, resonance 71) comes next, one filter for each voice. The voices are added together after their filters.
6. **Unison (61)** makes one key sound up to 8 voices, and every key you hold sounds its own voices.

What makes it louder

Control	What it does to the level
Amp level (75)	Every voice, every note
B level (24)	Adds the second oscillator to every voice
A level (27)	The first oscillator's share; lower it to make room for B
Resonance (71)	Raises the level of the sound around the cutoff
Unison voices (61)	One key sounds that many voices; their sum is louder than one voice, by an amount that depends on the key
Notes held	Every held note adds its voices to the sum
How hard you play	With Velocity > amp (76) above 0, harder keys are louder

Setting the level

1. Pick the densest thing the patch will play: the biggest chord, with unison, B and resonance at the settings you will use, and the hardest touch.
2. Play it. If it breaks up, lower **Amp level (75)** a few steps.
3. **Release the keys and play again.** A change to 75 or 76 takes effect at the next note you play. Notes that are already sounding keep the level they started with, so turning 75 down while a chord rings changes nothing you can hear until the next chord.
4. Repeat until the densest passage is clean, then save the patch, because 75 is part of the patch (see section 4).

Notes

- **Lower 75, not ENV1's peak (51) and sustain (53).** Those two shape the note. Turning them down to tame a chord changes how the note swells and holds as well as how loud it is. 75 changes only how loud the note is.
- Lowering 75 lowers single notes too. A patch set up for a big chord will be quiet when you play one note. Balance patches against each other at your mixer or amplifier, not with 75.

- Patches differ. A patch with unison, B and resonance may need a lower 75 than a plain one. Set each patch for the most it will be asked to do.

11. Parameter reference

This table is made from the instrument's own parameter table, so the numbers and ranges are the ones the panel shows and the printed card repeats.

- **Number** is the two digits you type after PARAM, and is the same number in the MIDI controller column and in a MIDI editor.
- **Shows** is what the display reads, not the stored byte. A detune shows 50 for no shift; the channel shows 1 to 16; key speed shows 1 to 99 with 50 as none. **0 to n** means the last entry of the patch's own list of tables (10) or sequences (11).
- **CC** is the MIDI controller that edits it, the same number, with 0 to 127 scaled onto the range shown. A dash means it has no CC: set it from the panel, or from a computer by SysEx (section 12).

Number	Name	Shows	CC	Notes
10	Wavetable	0 to n	-	the patch's tables, 0 first
11	Wave sequence	0 to n	-	the patch's sequences, 0 first
12	Wave position	0-99	12	start, 0 first entry to 99 last, any table
13	Wave wrap	0-1	13	0 clamp, 1 wrap
14	Velocity > wave	0-99	14	-
15	Key track > wave	0-99	15	-
17	LFO > wave	0-99	17	-
18	Sequence mode	0-4	18	0 off, 1 step, 2 per voice, 3 random, 4 LFO
19	Sequence reset gap	0-99	19	silence before restart, 0 never
20	Coarse A	38-62	20	semitones, 50 = 0
21	Fine A	29-71	21	1/256 octave (4.7 cents), 50 = 0
22	Coarse B	38-62	22	semitones, 50 = 0
23	Fine B	29-71	23	1/256 octave (4.7 cents), 50 = 0
24	Oscillator B level	0-99	24	0 = off; adds to A, so the voice gets louder
25	Bend range	0-12	25	whole semitones, full wheel; 2 is a whole tone, 12 an octave
26	Vintage	0-99	26	0 off; 1-99 more drift: each note-on gives every voice its own pitch and cutoff offset, up to +/-8 steps (+/-38 cents; +/-3.4 semitones of cutoff) at 99
27	Oscillator A level	0-99	27	99 = full, 0 = off
28	Oscillator B source	0-7	28	0 A's wave, 1 white noise, 2 pink, 3 scrape (tracking the key), 4 white, 5 pink, 6 scrape (fixed rate, C4), 7 sync (B is A's wave, hard-synced to A)
30	LFO rate	0-99	30	-
31	LFO > pitch	0-99	31	-
32	LFO shape	0-4	-	0 triangle, 1 ramp up, 2 ramp down, 3 square, 4 random

33	LFO sync	0-2	-	0 free-running, 1 restart at note-on, 2 MIDI clock (not built: runs as 0)
34	LFO > cutoff	0-99	34	-
35	LFO delay	0-99	35	-
36	Mod wheel depth	0-99	36	-
37	Pressure depth	0-99	37	channel aftertouch, 0 = off
38	Mod wheel destination	0-1	-	0 LFO depth, 1 wave position
39	Pressure destination	0-2	39	0 LFO depth, 1 wave position, 2 cutoff
40	ENV2 attack	0-99	40	-
41	ENV2 peak	0-99	41	-
42	ENV2 decay	0-99	42	-
43	ENV2 sustain	0-99	43	-
44	ENV2 release	0-99	44	-
45	Velocity > ENV2	0-99	45	0 = depth as set (72); 99 scales it by velocity / 127
47	ENV2 key speed	1-99	47	50 = none; 99 halves decay and release per octave above key 60
50	ENV1 attack	0-99	50	-
51	ENV1 peak	0-99	51	-
52	ENV1 decay	0-85	52	exponential
53	ENV1 sustain	0-99	53	-
54	ENV1 release	0-85	54	exponential
57	ENV1 key speed	1-99	57	50 = none; as 47
60	Voice mode	0-4	60	0 poly, 1 mono retrigger, 2 mono legato, 3 chord retrigger, 4 chord legato
61	Unison voices	1-8	61	per note; 1 = off
62	Unison spread	0-21	62	1/256 octave (4.7 cents) a step
63	Glide	0-99	63	0 off; 1-99 slower, an octave in 22 ms to 33 s
70	Filter cutoff	0-99	70	-
71	Resonance	0-99	71	high settings self-oscillate
72	ENV2 > cutoff	0-99	72	-
73	Key track > cutoff	0-99	73	-
74	Velocity > cutoff	0-99	74	-
75	Amp level	0-99	75	lower it if chords or unison distort
76	Velocity > amp	0-99	76	-
77	Wave > cutoff	1-99	77	bipolar, 50 = none: + opens the filter up the table
80	ENV3 attack	0-99	80	-
81	ENV3 peak	0-99	81	-
82	ENV3 decay	0-99	82	-
83	ENV3 sustain	0-99	83	-

84	ENV3 release	0-99	84	-
85	ENV3 destination	0-5	85	0 pitch A+B, 1 pitch A, 2 pitch B, 3 resonance, 4 LFO depth, 5 wave position
86	ENV3 depth	0-99	86	bipolar, 50 = none
87	ENV3 key speed	1-99	87	50 = none; as 47
88	ENV3 loop	0-1	-	0 = off; 1 = restarts from its attack at the end of its decay while the key is held
93	Save user waves	action	-	the changed ones, at the second digit
94	Restart sequence	action	-	at the second digit
95	MIDI channel	1-16	-	numbered as a DAW numbers them
96	Omni	0-1	-	0 off, 1 on
97	Master tune	29-71	-	saved with the channel and omni; 50 = none, 4.7 cents a step
98	Transpose	38-62	-	saved with the channel and omni; 50 = none, semitones

A number that is not in the table is not a parameter, and entering one shows EE.

12. MIDI

Channel and omni

Parameter **95** is the channel Phantasm receives on, shown 1 to 16. **96** is omni: 1 receives on every channel, 0 only on 95's. Both are saved with a patch save when they have changed (section 4).

Master tune and transpose

Parameter **97** tunes the whole instrument and **98** transposes it, for matching the Mirage to another instrument or a track. Both are 50 for none.

- **97, master tune,** is 29 to 71: a step is 1/256 of an octave (4.7 cents), up to 21 steps (about 100 cents) either way. 4.7 cents is coarse against a reference pitch; it is the pitch unit the engine has, the same as a patch's fine tune (A fine).
- **98, transpose,** is 38 to 62: whole semitones, 12 either way.
- Both are applied when a note starts, so a change is heard from the next note and a held note stays where it was. A mono legato slur takes the current values.
- Transpose moves the pitch and not the filter: the cutoff's key tracking and the envelopes' key speed follow the key you play, so a transposed patch keeps its brightness. Fixed-rate noise (28 = 4-6) is pitched from C4 whatever the key, so transpose does not move it; the master tune does.
- Neither moves the sequence, and neither is a patch parameter: both are saved with the channel and omni (section 4), not with a patch.
- They are not on a CC. The editor sets them by SysEx like 95 and 96.

What it responds to

Message	Effect
Note on and off	Plays, with velocity
Pitch bend	Bends every voice, by the range in 25
CC 1	The mod wheel (36 and 38 set what it does)
CC 64	Sustain pedal
Program change 0 to 99	Loads patch 00 to 99. A disk load of about two seconds
CC 120, 123, 124, 125	Release every voice (120, all sound off, and 123, all notes off, are the same here)
CC 121	Reset all controllers: the bend to centre, the mod wheel and channel pressure to 0, the sustain pedal up (held notes release). The patch is not touched
CC 12 to 31, 34 to 37, 39 to 63, 70 to 87	Edits the parameter with the same number (section 11)
Channel pressure	Aftertouch: depth 37, destination 39 (section 8)
Polyphonic pressure	Read and ignored

Every other controller keeps its usual meaning and is ignored, so a DAW's volume, pan and reverb sends do not change a patch. A CC for a number that has no parameter is dropped. A CC's 0 to 127 is scaled onto the parameter's own range, so a controller's centre value is 50 on a detune.

CC 121 is the cure for a bend or wheel left off-centre when a controller is unplugged mid-phrase. The Mirage's own wheels are read again only when one is moved, so a reset holds until you touch a wheel.

A CC edits a parameter the same way the panel does, and the display then shows that parameter. A program change received while the disk is busy is lost, and so is a note.

Editing from a computer

A computer can read and set every parameter by SysEx, dump and write the loaded patch, and select and save patches by number. phantasmeditor, a separate project, is the editor for it. The protocol is in docs/midi-protocol.md; this firmware speaks SysEx protocol 4.0, which the discovery request reports, and an editor written for an earlier one cannot dump or write a patch to it. The Mirage only answers: it never sends anything by itself.

A written patch goes to memory only. Nothing reaches the disk until a save, from the panel or by the editor, so an interrupted transfer cannot damage a patch. The display is not redrawn after a write to the loaded patch until you touch the panel.

Sending your own waves

Phantasm receives the MIDI Sample Dump Standard into the user area (section 5). From a sample editor or a dump program:

- Send **8-bit** samples of **256 points per wave**, so one wave is 256 samples and a longer sample is that many waves in a row. Sample number n, 0 to 127, goes to wave 384 plus n.
- It is **one way**. The Mirage sends nothing back, so no handshake: turn the sender's handshake or wait-for-ACK off, and leave 2 seconds after the header and 20 milliseconds between packets.
- 128 waves take about 35 seconds. The keys and MIDI notes are ignored until the dump ends, or for 2.6 seconds after the last packet.
- Waves are written whole or not at all. A bad packet abandons the dump and keeps the waves already written.
- Each sample keeps its top 8 bits, and a zero becomes 1. Send each wave as one clean cycle, centred and levelled by your editor: the instrument does not condition it.

- **The waves are in wave memory only.** Press PARAM 9 3 to save them (section 5).

Do not send other SysEx while a dump runs: any other SysEx message ends it.

13. Display codes and errors

What the display shows

Display	Meaning
Two digits	The patch number, or a parameter's number or value as you type
12, or 1.2	The version, from boot until you touch the panel
Ld	A patch is loading
Sd	A save is writing: a patch, or the user waves
EE	A parameter number that does not exist
E and a digit or letter	The disk could not be read or written. The character is the drive's last status; only the E carries a message
EP	A save was refused because the disk is write protected
EU	Action 93 was refused: the loaded disk has no user area
Er	Action 93 wrote a wave and read it back different

What to do

- **E and a character on a load.** The patch did not read. Try the load again. If it fails again, try another patch number, then restart. The instrument is playing a plain default sound, and **will refuse to save** until a patch loads well, so a read error cannot overwrite a good patch. A disk built for an earlier firmware does the same on every load: the patch was 72 bytes before SysEx protocol 4.0 and is 96 now, a patch sector from before is refused, and the disk has to be rebuilt.
- **E and a character on a save.** The write or its read-back failed. The patch may not be on the disk: load it again to see, and save again.
- **EP.** The disk is write protected. Remove the write protection of the image or the stick, restart, and save again.
- **EU.** The disk was not made for the user area. It holds fewer than 512 waves, and the user waves are not saved. Use a disk built for 512.
- **Er.** A user wave did not read back as written. That wave and the ones after it are still marked as changed. Press PARAM 9 3 again. If it persists, the disk or the drive is at fault.
- **The patch number changed after a failed save-as.** A save-as that fails on the disk (EP, or E and a character) still moves the displayed number to the number you typed, though the patch may not be there. Load the patch you meant to be on. A save-as refused after a failed load leaves the number where it was.
- **A MIDI write did not show on the display.** It reached the sound. Touch any panel key.
- **Notes are silent.** Check parameter 95 and omni 96, 75 (amp level), and that a user-wave dump or a disk transfer is not still running.
- **Nothing appears after boot.** Wait the full half-minute. If it still shows nothing, check the Gotek has the Phantasm image selected and restart. A bad image cannot hurt the Mirage: its boot ROM is untouched.

Starting over

Unsaved edits are lost on any load and at power off. Saved patches and saved user waves are on the disk. If the sound is wrong and you do not know why, load the patch again.